INCH-POUND
MIL-DTL-16377/79A
w/AMENDMENT 1
25 July 2022
SUPERSEDING
MIL-DTL-16377/79A
30 June 2014

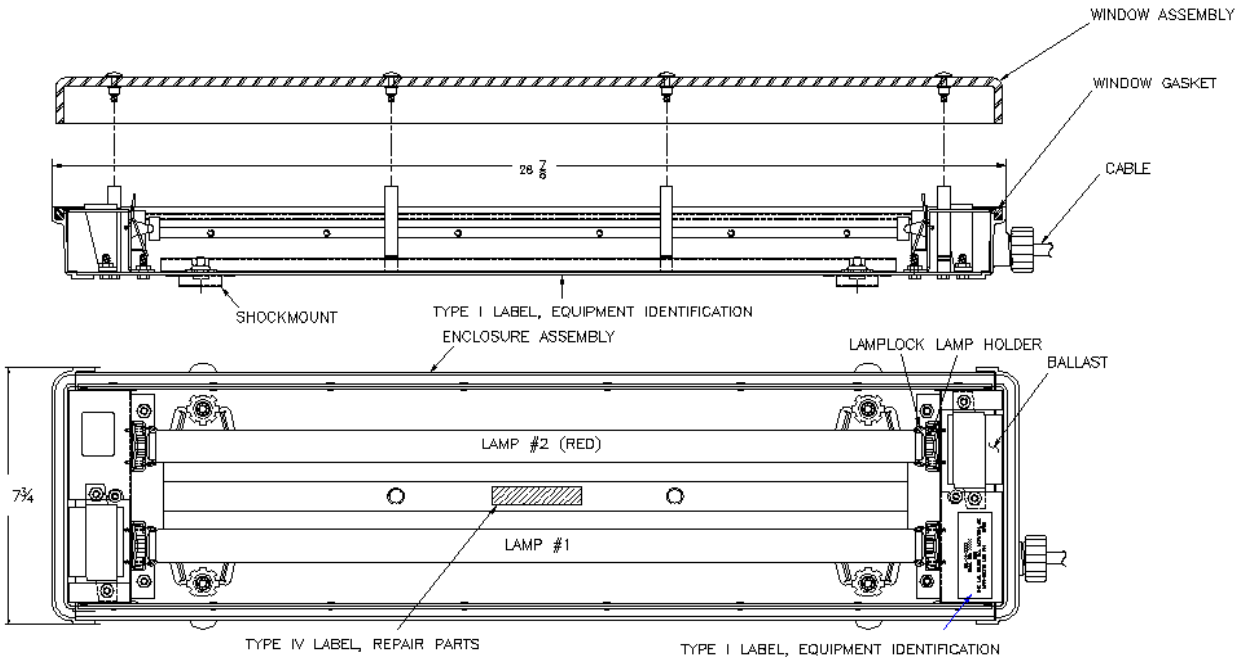
DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; T8 FLUORESCENT, GENERAL LIGHTING, 34 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 77.4T8, 77.5T8, 344.2T8, AND 344.3T8

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377. If specified by the acquisition activity, solid state lighting (SSL) fixtures designed and manufactured in accordance with MIL-DTL-16377/11 may be substituted for fixtures manufactured in accordance with this specification sheet.

All dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.



NOTE:

1. Symbols 77.5T8 and 344.3T8 use identical construction with flush mounting flange not shown

FIGURE 1. Typical view of symbols 77.4T8 and 344.2T8.

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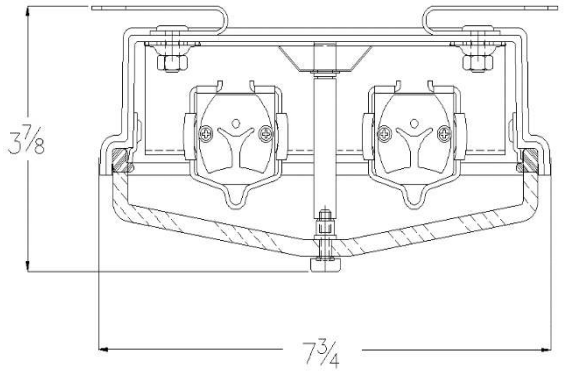
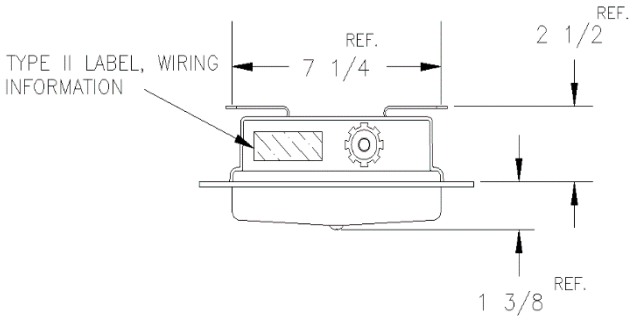


FIGURE 2. Typical section for symbols 77.4T8 and 344.2T8.



NOTE:

1. Type II label required for all fixtures.

FIGURE 3. Typical section for symbols 77.5T8 and 344.3T8.

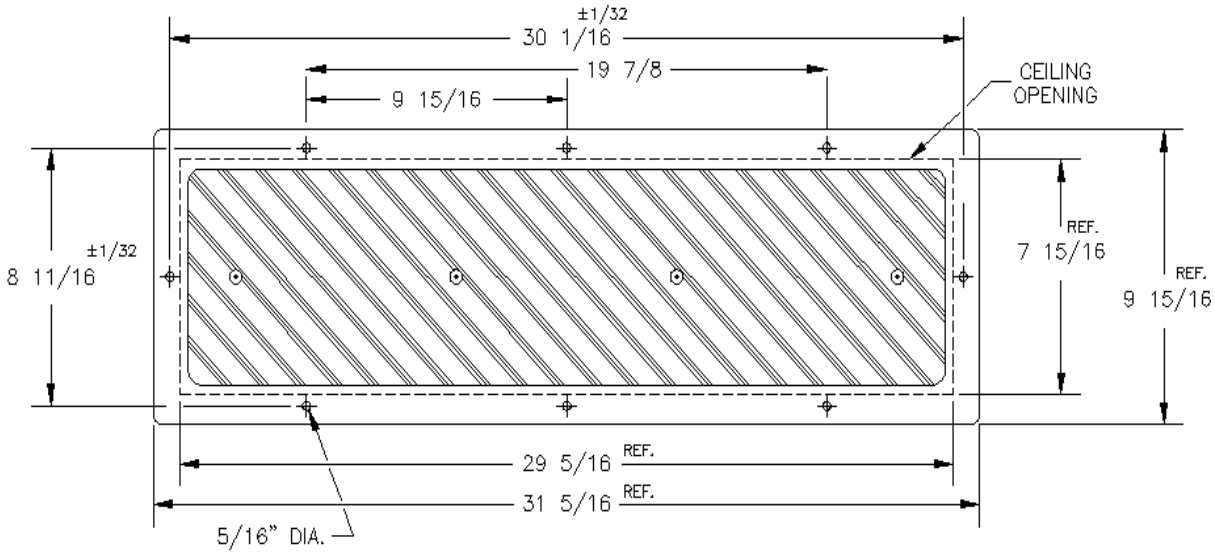


FIGURE 4. Flush mounting dimensions for symbols 77.5T8 and 344.3T8.

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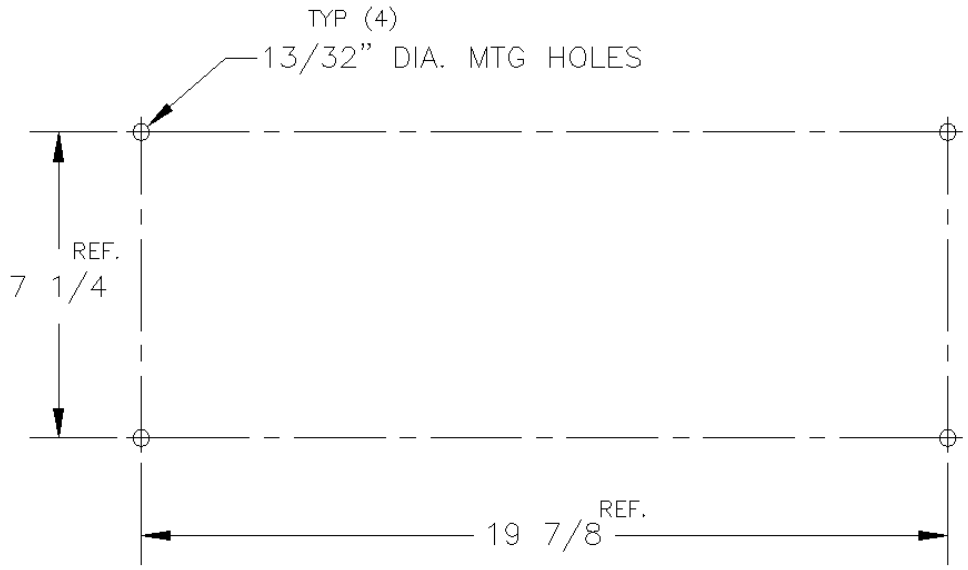


FIGURE 5. Standard mounting dimensions utilizing shock mounts for symbols 77.4T8 and 344.2T8.

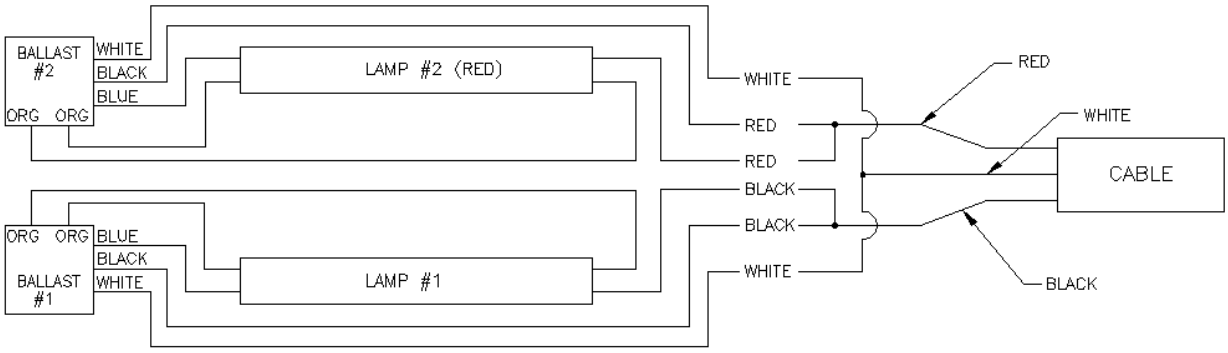


FIGURE 6. Typical wiring diagram.

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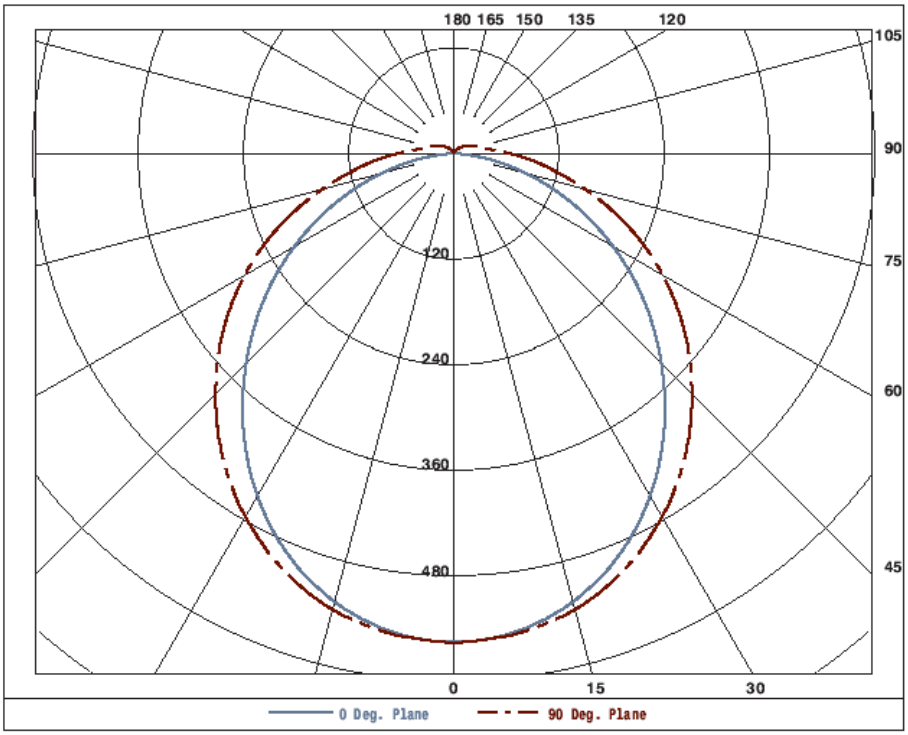


FIGURE 7. Candela distribution curves, white diffusing window, symbols 77.4T8 and 77.5T8.

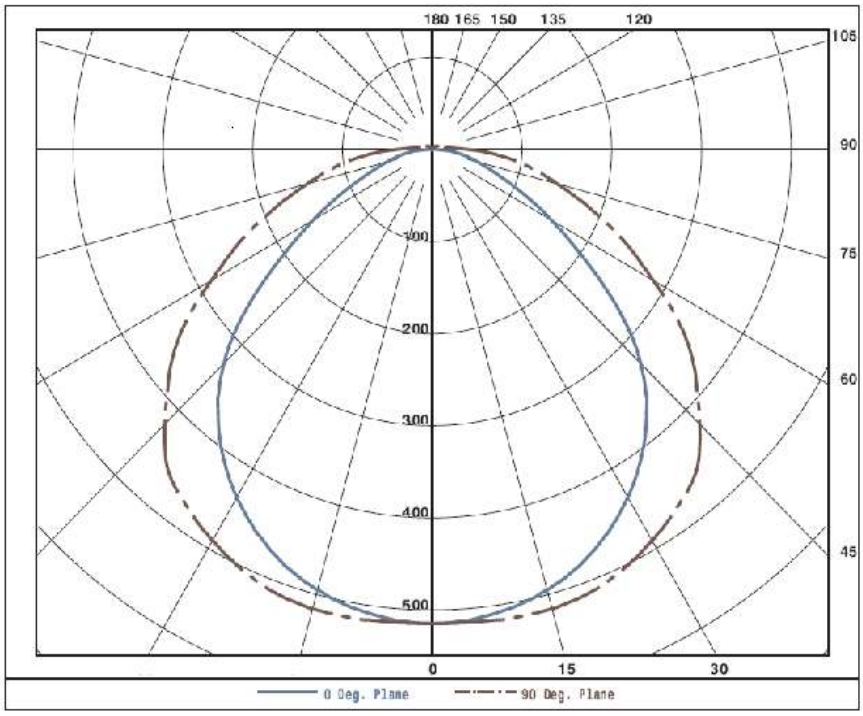


FIGURE 8. Candela distribution curves, prismatic window, symbols 344.2T8 and 344.3T8.

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TABLE I. Numerical candela distribution, white diffusing window, symbols 77.4T8 and 77.5T8.

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
0	556	556
5	552	552
15	528	537
25	480	501
35	413	449
45	335	385
55	253	314
65	167	236
75	85	155
85	18	90
90	4	65

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
95	4	46
105	3	30
115	2	21
125	2	14
135	1	8
145	0	4
155	0	0
165	0	0
175	0	0
180	0	0

TABLE II. Numerical candela distribution, prismatic window, symbols 344.2T8 and 344.3T8.

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
0	515	515
5	512	515
15	495	516
25	461	503
35	407	475
45	328	421
55	207	340
65	109	236
75	52	144
85	19	75
90	5	40

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
95	4	20
105	3	13
115	2	11
125	1	6
135	1	3
145	0	2
155	0	0
165	0	0
175	0	0
180	0	0

PHOTOMETRIC DATA:

Photometry reports based on LM-41 and other pertinent Illuminating Engineering Society (IES) procedures.

Plane:	0 degrees is parallel to lamps, 90 degrees is perpendicular to lamps.
Electrical values:	120 volts, 60 hertz, 36.1 watts input power.
Ambient temperature:	77±1.8 °F (25±1 °C).
Lamps:	GE F17T8-SP41-ECO, or equal (seasoned for at least 100 hours).
Lamps correction:	Lamps rated at 1,325 lumens.
Ballast:	Furnished with fixture, see MIL-DTL-16377/81 for specification.

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REQUIREMENTS:

CLASSIFICATION:

Type: I
Class: 2

SYMBOL NUMBERS:

77.4T8: Complete lighting fixture assembly, standard mounting with two 17-watt T8 cool white fluorescent lamp and white diffusing window assembly.
77.5T8: Complete lighting fixture assembly, flush mounting with two 17-watt T8 cool white fluorescent lamp and white diffusing window assembly.
344.2T8: Complete lighting fixture assembly, standard mounting with two 17-watt T8 cool white fluorescent lamp and clear and white prismatic window assembly.
344.3T8: Complete lighting fixture assembly, flush mounting with two 17-watt T8 cool white fluorescent lamp and clear and white prismatic window assembly.

DESIGN:

Electrical rating: 34 watts, 120 volts, 60 hertz.
Dimension and configurations: See figures 1, 2, and 3.
Photometric data: See figures 7 and 8 and tables I and II.
Wiring diagram: See figure 6 for typical wiring.
Ballast: See MIL-DTL-16377/81 for specification.
Mounting dimensions:
Symbols 77.4T8 and 344.2T8: Standard mounting (surface), see figure 5.
Symbols 77.5T8 and 344.3T8: Flush mounting (recessed), see figure 4.
Window assembly:
Symbols 77.4T8 and 77.5T8: White window, see table III and figure 27 of MIL-DTL-16377.
Symbols 344.2T8 and 344.3T8: Prismatic window, see table III and figure 27 of MIL-DTL-16377.
Enclosure: Watertight, see 3.5.11 of MIL-DTL-16377.
Cable: 10 feet (minimum) of Type LSTHOF-3 in accordance with MIL-DTL-24643/3.
Weight:
Symbols 77.4T8 and 344.2T8 Not to exceed 12 pounds.
Symbols 77.5T8 and 344.3T8 Not to exceed 13 pounds 13 ounces.
Designation and marking: The following labels shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figures 1 and 3:

a. Two Type I labels, equipment identification, reference MIL-DTL-16377 figure 11; shall include manufacturer and address, one located on back and other located on an inner nonreflecting surface.
b. One Type II label, circuit wiring information, reference MIL-DTL-16377 figure 12; located on outside surface adjacent to cable entrance.
c. One Type IV label, repair parts, reference MIL-DTL-16377 figure 14; information to be shown shall be “USE THE FOLLOWING REPAIR PARTS ONLY:” followed by a list of all standard parts shown in table III, located on an inner nonreflecting surface.

Standard parts: The standard parts shown in table III shall be used in the construction of the fixture specified herein.

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EXAMINATIONS AND TESTS:

Examinations and tests shall be as specified in MIL-DTL-16377 and the following:

First article testing/inspection:
Complete fixtures: See table IV.
Window assemblies: See table V.
Conformance:
Comparison inspection: Same as first article inspection.
Conformance inspection: See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of military part numbers and general information is shown in table VI.

TABLE III. Standard parts.

Name	Military P/N or Specification	National Stock Number (NSN)	Remarks
Lamp	Commercial identification F17T8/CW	6240-01-391-6027	
Lamplock	M16377/40-003		
Lampholder	UL 542	6250-00-284-0481	Commercial
Ballast	M16377/81-001		Integrated starter type
High-temperature ballast	M16377/81-002		High-temperature applications
High-temperature starter	FS-4	6250-00-179-2354	High-temperature applications
Startersocket	UL 496	6250-00-295-2738	High-temperature applications
Shock mount	M16377/41-008	6210-01-393-4302	0.067 steel, zinc plated
Window assembly, diffusing	M16377/11-001	6210-00-399-7035	1/
Window assembly, prismatic	M16377/11-003	6210-01-169-5997	1/
Filter, 17-watt T8 fluorescent lamp (Dark adaptation)	M16377/42-010	6210-01-313-8169	Red transparent
Filter, 17-watt T8 fluorescent lamp (1% transmission gray)	M16377/42-011		
Filter, 17-watt T8 fluorescent lamp (2.5% transmission gray)	M16377/42-012		
Filter, 17-watt T8 fluorescent lamp (Amber)	M16377/42-015		
Filter, 17-watt T8 fluorescent lamp (Broad band blue)	M16377/42-016		

NOTE:
1/ Window assemblies (diffusing and prismatic) are the same as T12 lamp fixtures (MIL-DTL-16377/11).

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TABLE IV. First article tests and comparison inspection for complete fixtures. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	2/
Voltage transient spike	3.6.18.2	4.8.23	3/
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	4/
Insulation resistance	3.6.16	4.8.3	4/
Enclosure effectiveness	3.5.11	4.8.14.4	4/ Watertight
Shock	3.5.12	4.8.7	4/
Vibration	3.5.13	4.8.8	4/
Enclosure effectiveness	3.5.11	4.8.14.4	4/ Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	4/
Insulation resistance	3.6.16	4.8.3	4/
Noise	3.5.14	4.8.9	
Continuity of grounding	3.6.13.4	4.8.12	
Light output	3.7.1	4.8.6.1	
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	

NOTES:
1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.
2/ If less, then 1 KVA exception not applicable for input waveform total harmonic distortion.
3/ Solid state components shall be subject to 250 positive spikes and 250 negative spikes (500 total spikes), one spike per minute, using testing method identified in MIL-STD-1399-300-1.
4/ Tests shall be performed in the order specified.

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TABLE V. First article tests and comparison inspection for window assemblies. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14.4	2/ Watertight
Light output	3.7.1	4.8.6.1	Candela distribution

NOTES:
1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.
2/ Tests shall be performed in the order specified.

TABLE VI. Complete fixture part numbers.

Military Part Number M16377/79-	Item Name	National Stock Number (NSN)
77.4T8	Complete fixture, Symbol 77.4T8	6210-01-593-2386
77.5T8	Complete fixture, Symbol 77.5T8	
344.2T8	Complete fixture, Symbol 344.2T8	
344.3T8	Complete fixture, Symbol 344.3T8	

AMENDMENT NOTATIONS: The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army – CR4
Navy – SH
Air Force – 03

Review activities:
Navy – AS, CG
DLA – GS2

Civil agency:
GSA – FAS

Preparing activity:
Navy – SH
(Project 6210-2022-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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